Generalizing Brown's Reduced McCallum Projection to Elimination Ideals

Notes for the `mccalum` formalization project

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Abstract

The repository `mccalum` contains a Lean 4 proof of a generalization of McCallum's projection theorem for cylindrical algebraic decomposition (CAD): discriminants and resultants are replaced by arbitrary nonzero elements of the corresponding elimination ideals. These notes investigate whether the same generalization applies to Brown's reduced McCallum projection [1], specifically to Brown's Theorem 3.1, which removes all non-leading coefficients from the projection. We show that the verbatim generalization is *false*, by an explicit counterexample, and that the failure is structural: the affine elimination ideal $\langle f, f_z\rangle \cap \mathbb{R}[\bar{x}]$ is blind to roots at infinity, while degree-invariance — the conclusion of Brown's theorem — is a statement about infinity. We then prove a corrected generalization: it suffices to require the witness polynomial to lie in the elimination ideals of both f *and its degree-n reverse*, a condition equivalent to membership in the projective (saturated) elimination ideal of the homogenization, satisfied by the discriminant, and stable under the shift-and-reverse transformations used in Brown's proof. Finally, we describe how a single Bézout certificate for f, as produced by an external solver, can be transformed mechanically into the second required certificate at the cost of a factor $\mathrm{lc}(f)^M$ — a factor that adds no new projection factors and no cost to the CAD algorithm — and we assess the difficulty of formalizing the corrected theorem in Lean on top of the existing development.

1 Setting and notation

Throughout, $R := \mathbb{R}[x_1, \ldots, x_k] = \mathbb{R}[\bar{x}]$ and $f \in R[z]$ is a polynomial of positive degree $n := \deg_z f$, written $f = \sum_{i=0}^{n} a_i z^i$ with $a_i \in R$ and $a_n \neq 0$. We write $f_z := \partial f/\partial z$, and $\mathrm{lc}\, f := a_n$ for the leading coefficient. For $q \in \mathbb{R}^k$ we write $f(q, \cdot) \in \mathbb{R}[z]$ for the specialization (in the Lean development, `specialize f q`). The central object of these notes is the *elimination ideal*

$$\mathcal{E}(f) \;:=\; \langle f,\, f_z\rangle \cap R,$$

the ideal of R consisting of polynomials in $\bar{x}$ alone that can be written as $Af + Bf_z$ with $A, B \in R[z]$.

For a nonzero polynomial $g \in R$ and $p \in \mathbb{R}^k$, $\mathrm{ord}_p\, g$ denotes the order of vanishing of g at p (the least total degree of a nonvanishing partial derivative). g is *order-invariant* on $S \subseteq \mathbb{R}^k$ if $p \mapsto \mathrm{ord}_p\, g$ is constant on S, and *sign-invariant* on S if $p \mapsto \mathrm{sign}\, g(p) \in \{-1, 0, +1\}$ is constant on S. Note that "identically zero on S" is sign-invariant. Analytic submanifolds of $\mathbb{R}^k$ and analytic delineability are as in McCallum [2] and as formalized in `Mccalum/Submanifold.lean` and `Mccalum/Delineability.lean`; we emphasize that the formalized notion of delineability (`AnalyticDelineable`) includes *constant multiplicities*: the real roots of $f(q, \cdot)$ over S are given by finitely many analytic sections $\theta_1 < \cdots < \theta_s$ on S, the $\theta_j(q)$ exhaust the real roots of $f(q, \cdot)$ for every $q \in S$, and the multiplicity of $\theta_j(q)$ as a root of $f(q, \cdot)$ is a constant $m_j \geq 1$.

The starting point is the following theorem, already proved in the repository; it generalizes McCallum's Theorem 2 [2] (= Theorem 2.1 of [1]) by replacing the discriminant with an arbitrary nonzero element of $\mathcal{E}(f)$. In Lean it is `lifting_theorem_generalized`, in `Mccalum/Generalized/Projection.lean`.

Theorem 1.1 (Generalized lifting theorem, Theorem 3.2.1′; proved in Lean)**.** *Let S be a connected analytic submanifold of $\mathbb{R}^k$, and let $f \in R[z]$ be degree-invariant on S with $f(q,\cdot) \not\equiv 0$ for every $q \in S$. Suppose there is $P \in R$, $P \neq 0$, with $P \in \mathcal{E}(f)$, such that P is order-invariant on S. Then f is analytically delineable on S and is order-invariant in each f-section over S.*

2 Brown's theorem and its naive generalization

McCallum's projection of f contains *all* coefficients $a_n, \dots, a_0$ (to secure degree-invariance on cells), together with the discriminant and the pairwise resultants. Brown's contribution [1, Theorem 3.1] is that the non-leading coefficients are redundant:

Theorem 2.1 (Brown, 2001)**.** *Let $f \in \mathbb{R}[\bar{x}][z]$ have positive degree n in z, with discriminant $D(\bar{x}) \neq 0$. Let S be a connected analytic submanifold of $\mathbb{R}^k$ on which D is order-invariant and $\operatorname{lc} f$ is sign-invariant, and such that f vanishes identically at no point of S. Then f is degree-invariant on S.*

Degree-invariance is exactly the hypothesis that Theorem 1.1 needs; combining the two yields Brown's *reduced* projection: leading coefficients, discriminants, and resultants only.

Given the repository's generalization of McCallum's theorem, the natural candidate statement is:

Conjecture 2.2 (Naive generalization — **false**)**.** *Let $f \in \mathbb{R}[\bar{x}][z]$ have positive degree n in z. Let $P \in \mathcal{E}(f)$, $P \neq 0$. Let S be a connected analytic submanifold of $\mathbb{R}^k$ on which P is order-invariant and $\operatorname{lc} f$ is sign-invariant, and such that f vanishes identically at no point of S. Then f is degree-invariant on S.*

Note that the hypothesis already implies $D \neq 0$: if f had a repeated factor over $\mathbb{R}(\bar{x})$, then $\gcd(f, f_z)$ would have positive z-degree and divide every element of $\langle f, f_z \rangle$ in $\mathbb{R}(\bar{x})[z]$, forcing $\mathcal{E}(f) = 0$. So Conjecture 2.2 genuinely only relaxes the *order-invariance* hypothesis from D to P, in exact analogy with the proved Theorem 1.1.

3 A counterexample

Proposition 3.1. *Conjecture 2.2 is false. Let $k = 2$, $\bar{x} = (x, y)$, and*

$$f \;=\; -xy^2z^2 + yz + 1, \qquad n = 2, \qquad \operatorname{lc} f = -xy^2,$$

let $S = \{(0,t) : t \in \mathbb{R}\}$ be the y-axis, and let $P = 1+4x$. Then all hypotheses of Conjecture 2.2 hold, but f is not degree-invariant on S.

Proof. First, $P \in \mathcal{E}(f)$, with the explicit certificate

$$1 + 4x \;=\; \bigl(1 - 2xyz + 4x\bigr) \cdot f \;+\; \bigl(-z\,(1 + 2x - xyz)\bigr) \cdot f_z, \tag{1}$$

which is verified by direct expansion. (It can be found by noting $2f - zf_z = yz + 2$ and $1 + 4x = -f + (yz + 2)(1 - x(yz - 2))$.) Clearly $P \neq 0$.

S is a connected analytic submanifold of $\mathbb{R}^2$ (the regular zero set of $(x, y) \mapsto x$). On S we have $P(0, t) = 1$, so P is nowhere zero on S and hence order-invariant (constant order 0).

The leading coefficient $-xy^2$ vanishes identically on S, hence is sign-invariant (constant sign 0). Finally $f(0,t,z) = tz + 1$ is never the zero polynomial.

Yet $\deg_z f(0,t,\cdot) = \deg_z(tz+1)$ equals 1 for $t \neq 0$ and 0 for $t = 0$: the degree jumps at the origin. □

Remark 3.2 (Where the degree-drop hides from $\mathcal{E}(f)$)**.** Over $\mathbb{C}$, the common zeros of (f, f_z) are easily computed: $f_z = y(1-2xyz)$, and on $y \neq 0$, $1 - 2xyz = 0$ forces $yz = -2$ and then $x = -1/4$. Thus

$$V_{\mathbb{C}}(f, f_z) \;=\; \{\, (-\tfrac{1}{4},\ t,\ -2/t) : t \in \mathbb{C}^{\times} \},$$

whose projection to (x,y)-space has Zariski closure $\{x = -1/4\}$. By elimination theory, $V_{\mathbb{C}}(\mathcal{E}(f)) = \{x = -1/4\}$, which is disjoint from (the complexification of) S. This is why an element of $\mathcal{E}(f)$, such as $1+4x$, can be a unit along S: the elimination ideal of (f, f_z) records only *finite* multiple roots, and there are none above S. The degree drop at the origin is a collision of roots *at infinity* ($tz+1$ has its root $-1/t \to \infty$ as $t \to 0$), invisible to $\langle f, f_z\rangle \cap R$.

The discriminant, in contrast, does see infinity: $D = y^2(1+4x)$, and the formal degree-n discriminant always vanishes on the locus $\{a_n = a_{n-1} = 0\}$ where the degree drops by at least two. On S, D has order 0 at $(0,t)$, $t \neq 0$, and order 2 at the origin — it is *not* order-invariant, so Brown's actual hypotheses correctly exclude this S. In Brown's proof, the "sees infinity" property of D is used in exactly one place: his Lemma 8.1, $\operatorname{disc} f = \operatorname{disc}(\operatorname{rev} f)$, which transports the order-invariance hypothesis to the reversed auxiliary polynomial. That is the property an arbitrary $P \in \mathcal{E}(f)$ lacks, and the corrected generalization restores it by hypothesis.

4 The corrected generalization

4.1 Reversal and homogenization

Definition 4.1. For $g = \sum_{i=0}^{m} c_i z^i \in R[z]$ with $\deg g \le m$, the *degree-m reverse* is $\operatorname{rev}_m g := \sum_{i=0}^{m} c_i z^{m-i} = z^m\, g(1/z)$. (In Mathlib: `Polynomial.reflect m g`; for $m = \deg g$ this is `Polynomial.reverse`.) For f as in Section 1 we write $\operatorname{rev} f := \operatorname{rev}_n f = a_0 z^n + a_1 z^{n-1} + \cdots + a_n$. Note the constant coefficient of $\operatorname{rev} f$ is $a_n = \operatorname{lc} f$, and rev_m is multiplicative in the sense $\operatorname{rev}_{d+e}(QR) = \operatorname{rev}_d(Q)\ \operatorname{rev}_e(R)$ whenever $\deg Q \le d$, $\deg R \le e$ (Mathlib's `reflect_mul`).

We emphasize that the relevant second ideal is $\langle \operatorname{rev} f,\ (\operatorname{rev} f)' \rangle$, with the derivative taken *after* reversing; reversal and d/dz do not commute. They are related by an Euler-type identity:

Lemma 4.2. *For $f \in R[z]$ with $\deg f \le n$,*

$$\operatorname{rev}_{n-1}(f_z) \;=\; n \cdot \operatorname{rev}_n f \;-\; z\,(\operatorname{rev}_n f)'.$$

Proof. Both sides equal $\sum_{i=0}^{n} i\, a_i z^{n-i}$: the left side because $f_z = \sum_i i a_i z^{i-1}$ reverses (at degree $n-1$) to $\sum_i i a_i z^{(n-1)-(i-1)}$; the right side because $(\operatorname{rev}_n f)' = \sum_i (n-i) a_i z^{n-i-1}$, so $n\,\operatorname{rev}_n f - z(\operatorname{rev}_n f)' = \sum_i (n-(n-i)) a_i z^{n-i}$. □

Definition 4.3. For $\varphi = \sum_{i=0}^{N} c_i z^i \in R[z]$ with $\deg \varphi \le N$, the *formal degree-N homogenization* is $H_N(\varphi) := \sum_{i=0}^{N} c_i Z^i W^{N-i} \in R[Z,W]$, homogeneous of degree N in (Z,W). Let $\varepsilon : R[Z,W] \to R[z]$ be the substitution $Z \mapsto z$, $W \mapsto 1$; then $\varepsilon(H_N \varphi) = \varphi$, and $H_N \circ \varepsilon$ is the identity on homogeneous polynomials of degree N. For homogeneous Φ set $J(\Phi) := \langle \Phi,\ \Phi_Z,\ \Phi_W \rangle \subseteq R[Z,W]$.

Lemma 4.4. *Let $\varphi \in R[z]$ with $\deg \varphi \le N$, $\Phi := H_N(\varphi)$, and $P \in R$.*

(a) $\Phi_Z = H_{N-1}(\varphi')$*, and by Euler's relation* $N\Phi = Z\Phi_Z + W\Phi_W$ *we get* $\varepsilon(\Phi_W) = N\varphi - z\varphi' \in \langle \varphi, \varphi' \rangle$*. Consequently, for* any *homogeneous* Φ *of degree* N*:*

$$\varepsilon\big(J(\Phi)\big) \subseteq \langle\, \varepsilon(\Phi),\ \varepsilon(\Phi)'\,\rangle.$$

In particular, if $P \cdot W^m \in J(\Phi)$ *for some* m*, then* $P \in \langle \varepsilon(\Phi), \varepsilon(\Phi)' \rangle$*.*

(b) Conversely, if $P \in \langle \varphi, \varphi' \rangle$*, say* $P = A\varphi + B\varphi'$ *with* $\deg A \le a$*,* $\deg B \le b$*, then*

$$P \cdot W^M \ \in\ \langle \Phi,\ \Phi_Z \rangle \ \subseteq\ J(\Phi), \qquad M := \max(a+N,\ b+N-1),$$

by homogenizing the identity term by term (H *is multiplicative on matching formal degrees, and* $H_{N-1}(\varphi') = \Phi_Z$*).*

(c) (Swap.) Let σ *be the substitution* $Z \leftrightarrow W$*. Then* $\sigma(H_N\varphi) = H_N(\operatorname{rev}_N \varphi)$*, and* $\sigma(J(\Phi)) = J(\sigma\Phi)$ *for every homogeneous* Φ*.*

(d) (Linear substitutions.) For invertible $T \in \mathrm{GL}_2(\mathbb{R})$ *acting by substitution* $\sigma_T(\Phi)(Z,W) := \Phi(T(Z,W))$*, one has* $\sigma_T(J(\Phi)) = J(\sigma_T\Phi)$ *(chain rule: the pair* (Φ_Z, Φ_W) *transforms by the invertible matrix* T^{T}*), and* $\sigma_T(\langle Z, W\rangle) = \langle Z, W \rangle$*.*

Proof. All items are direct computations with the definitions. For (a), note that $\Phi_Z = \sum_i i c_i Z^{i-1} W^{N-i}$ dehomogenizes to φ', and that ε is a ring homomorphism, so $\varepsilon(J(\Phi))$ lands in the ideal generated by the ε-images of the three generators, which by the two identities equals $\langle \varepsilon\Phi, (\varepsilon\Phi)' \rangle$. Item (a) holds for any homogeneous Φ, including those with vanishing "leading" coefficients — this is important below, where reversals may have degree $< n$. $\square$

4.2 The transfer lemma

Brown's proof works pointwise: at $p \in S$ it chooses $\gamma \in \mathbb{R}$ with $f(p,\gamma) \neq 0$ and passes to the auxiliary polynomial

$$f^*_\gamma \ :=\ \operatorname{rev}_n\big(f(\bar{x},\ z+\gamma)\big) \ =\ z^n\, f\Big(\bar{x},\ \tfrac{1}{z} + \gamma\Big),$$

whose roots are $1/(\rho - \gamma)$ for ρ the roots of f, and whose formal leading coefficient is $f(\bar{x}, \gamma)$. The corrected hypothesis must therefore put P in $\mathcal{E}(f^*_\gamma)$ for the (point-dependent!) shifts γ used in the proof. The following lemma shows that the two fixed, γ-free memberships suffice.

Lemma 4.5 (Transfer)**.** *Let* $f \in R[z]$*,* $\deg_z f = n \ge 1$*, and* $P \in R$*. Suppose*

$$P \in \langle f,\ f_z \rangle \qquad \textit{and} \qquad P \in \langle \operatorname{rev} f,\ (\operatorname{rev} f)' \rangle.$$

Then for every $\gamma \in \mathbb{R}$*,* $P \in \langle\, f^*_\gamma,\ (f^*_\gamma)'\,\rangle$*.*

Proof. Let $F := H_n(f)$ and $J := J(F)$. By Lemma 4.4(b), $P\,W^{M_1} \in J$ for some M_1. Applying Lemma 4.4(b) to $\operatorname{rev} f$ (which has degree $\le n$) and then the swap (c),

$$P\,W^{M_2} \in J\big(H_n(\operatorname{rev} f)\big) = J(\sigma F) = \sigma(J) \quad\Longrightarrow\quad P\,Z^{M_2} \in J.$$

With $M := \max(M_1, M_2)$, every monomial $Z^i W^j$ with $i + j = 2M$ has $i \ge M$ or $j \ge M$, hence

$$P \cdot \langle Z, W \rangle^{2M} \ \subseteq\ J. \tag{2}$$

Now fix γ and let T be the (invertible) substitution $Z \mapsto W + \gamma Z$, $W \mapsto Z$. A homogeneous polynomial of degree n is determined by its dehomogenization, and $\varepsilon(\sigma_T F) = \varepsilon\big(F(W +$

$\gamma Z,\ Z))$ evaluated as follows: $F(Z+\gamma W, W)$ dehomogenizes to $f(z+\gamma)$, so $F(Z+\gamma W, W) = H_n(f(z+\gamma))$, and applying the swap (Lemma 4.4(c)) gives

$$\sigma_T F \;=\; F(W+\gamma Z,\ Z) \;=\; \sigma\big(H_n(f(z+\gamma))\big) \;=\; H_n\big(\mathrm{rev}_n(f(z+\gamma))\big) \;=\; H_n(f^*_\gamma).$$

Applying the automorphism σ_T to (2) and using Lemma 4.4(d),

$$P\cdot\langle Z, W\rangle^{2M} \;\subseteq\; \sigma_T(J) \;=\; J(H_n(f^*_\gamma)),$$

in particular $P\,W^{2M} \in J(H_n(f^*_\gamma))$, and Lemma 4.4(a) yields $P \in \langle f^*_\gamma, (f^*_\gamma)'\rangle$. ☐

Remark 4.6. Display (2) identifies the corrected hypothesis intrinsically: P lies in the *$\langle Z, W\rangle$-saturated (projective) elimination ideal* of $J(F) = \langle F, F_Z, F_W\rangle$. Its variety is the *closed* image of the projective incidence variety $\{F = F_Z = F_W = 0\} \subseteq \mathbb{R}^k \times \mathbb{P}^1$, which contains both the finite multiple-root locus and the degree-drop-≥ 2 locus $\{a_n = a_{n-1} = 0\}$ (the point $[1:0]$ at infinity gives $F = a_n$, $F_W = a_{n-1}$ there). This is the smallest GL_2-invariant correction of $\mathcal{E}(f)$, so requiring anything less invites counterexamples of the same flavor as Proposition 3.1.

4.3 The theorem

We also need Brown's local refinement lemma [1, Lemma 8.2], an easy consequence of the straightening chart already formalized (`IsAnalyticSubmanifold.straightening_chart`):

Lemma 4.7 (Refinement)**.** *Let S be a connected analytic submanifold of $\mathbb{R}^k$, $p \in S$, and N_0 an open neighborhood of p. There is an open N with $p \in N \subseteq N_0$ such that $S \cap N$ is a connected analytic submanifold of $\mathbb{R}^k$.*

Proof sketch. Take a straightening chart Φ at p mapping S locally onto $(\mathbb{R}^s \times \{0\}) \cap \Phi(\cdot)$, shrink its source to lie in N_0, and pull back a small open ball B around $\Phi(p)$; then $\Phi(S \cap N) = B \cap (\mathbb{R}^s \times \{0\})$ is convex, hence connected, and the submanifold property is local. (The same argument already appears inline in `Mccalum/Generalized/Delineable.lean` to establish preconnectedness of $S \cap U$.) ☐

Theorem 4.8 (Generalized Brown, Theorem 3.1$'$)**.** *Let $f \in \mathbb{R}[\bar{x}][z]$ have positive degree n in z. Let $P \in R$, $P \neq 0$, with*

$$P \in \langle f,\ f_z\rangle \cap R \qquad \textit{and} \qquad P \in \langle \mathrm{rev}\, f,\ (\mathrm{rev}\, f)'\rangle \cap R.$$

Let S be a connected analytic submanifold of $\mathbb{R}^k$ on which P is order-invariant and $\mathrm{lc}\, f$ is sign-invariant, and such that f vanishes identically at no point of S. Then f is degree-invariant on S.

Proof. By sign-invariance of $a_n = \mathrm{lc}\, f$ on S, either a_n vanishes nowhere on S — and then $\deg f(q,\cdot) = n$ for all $q \in S$ and we are done — or $a_n \equiv 0$ on S; assume the latter, so $\deg f(q,\cdot) \leq n-1$ on S. Since a locally constant function on a connected space is constant, it suffices to show that $q \mapsto \deg f(q,\cdot)$ is constant near each $p \in S$ (within S).

Fix $p \in S$. Since $f(p,\cdot) \not\equiv 0$, choose $\gamma \in \mathbb{R}$ with $f(p,\gamma) \neq 0$, and an open $N_0 \ni p$ with $f(q,\gamma) \neq 0$ for all $q \in N_0$. Put $\bar{f} := f(\bar{x}, z+\gamma)$ and $f^* := f^*_\gamma = \mathrm{rev}_n \bar{f}$, with coefficients $f^* = \sum_{i=0}^n \bar{a}_i z^{n-i}$ where $\bar{f} = \sum_i \bar{a}_i z^i$. We record:

1. *f^* is degree-invariant on N_0, of degree n:* its formal leading coefficient is $\bar{a}_0 = \bar{f}(\bar{x}, 0) = f(\bar{x}, \gamma)$, which is nonzero at every point of N_0. In particular $f^*(q,\cdot) \not\equiv 0$ on N_0.

(2) *Multiplicity of* 0 *records the degree drop:* for $q \in N_0$, the multiplicity of 0 as a root of $f^*(q,\cdot)$ is the trailing degree

$$\operatorname{mult}_0 f^*(q,\cdot) \;=\; n-\max\{i : \bar{a}_i(q)\neq 0\} \;=\; n-\deg \bar{f}(q,\cdot) \;=\; n-\deg f(q,\cdot),$$

the last step because shifting z preserves degree. Moreover the constant coefficient of $f^*(q,\cdot)$ is $\bar{a}_n(q)=a_n(q)$, so for $q\in S$ (where $a_n\equiv 0$), 0 is a root of $f^*(q,\cdot)$, of multiplicity $n-\deg f(q,\cdot)\geq 1$.

(3) *Witness:* by Lemma 4.5, $P\in\langle f^*,(f^*)_z\rangle\cap R$, $P\neq 0$, and P is order-invariant on every subset of S.

By Lemma 4.7, pick $N\subseteq N_0$ with $p\in N$ and $S\cap N$ a connected analytic submanifold. By (1)–(3), all hypotheses of Theorem 1.1 hold for f^* on $S\cap N$, so f^* is analytically delineable on $S\cap N$: there are analytic sections $\theta_1<\cdots<\theta_s$ on $S\cap N$ exhausting the real roots of $f^*(q,\cdot)$, with constant multiplicities $m_1,\ldots,m_s\geq 1$.

By (2), 0 is a real root of $f^*(p,\cdot)$, so there is exactly one index i with $\theta_i(p)=0$ (the $\theta_j(p)$ are pairwise distinct). By continuity of the finitely many sections, shrink N to an open $N'\ni p$ such that $\theta_j\neq 0$ on $S\cap N'$ for all $j\neq i$. Now let $q\in S\cap N'$ be arbitrary: by (2), 0 is a real root of $f^*(q,\cdot)$, hence $0=\theta_j(q)$ for some j, hence $j=i$; therefore

$$n-\deg f(q,\cdot) \;=\; \operatorname{mult}_0 f^*(q,\cdot) \;=\; \operatorname{mult}_{\theta_i(q)} f^*(q,\cdot) \;=\; m_i,$$

so $\deg f(q,\cdot)=n-m_i$ for every $q\in S\cap N'$. This is the desired local constancy. $\square$

Remark 4.9 (Brown's theorem is the special case $P=D$)**.** The formal degree-n discriminant satisfies both memberships. First, $\operatorname{disc}_n(\mathfrak{f})\in\langle\mathfrak{f},\mathfrak{f}'\rangle$ holds for the generic polynomial $\mathfrak{f}=\sum_i \mathfrak{a}_i z^i$ over $\mathbb{Z}[\mathfrak{a}_0,\ldots,\mathfrak{a}_n]$ (a classical fact, cf. [3]; it is also how the repository recovers Theorem 3.2.3 from 3.2.3′), and hence for every specialization. Second, Brown's Lemma 8.1 gives $\operatorname{disc}_n(\mathfrak{f})=\operatorname{disc}_n(\operatorname{rev}_n\mathfrak{f})$ whenever $\mathfrak{a}_0\mathfrak{a}_n\neq 0$; since both sides are polynomials in the $\mathfrak{a}_i$ and agree on a Zariski-dense set, they are equal in $\mathbb{Z}[\mathfrak{a}_0,\ldots,\mathfrak{a}_n]$, so $D=\operatorname{disc}_n f=\operatorname{disc}_n(\operatorname{rev} f)\in\langle\operatorname{rev} f,(\operatorname{rev} f)'\rangle$ by the first fact applied to $\operatorname{rev} f$. Thus Theorem 4.8 strictly subsumes Theorem 2.1.

Remark 4.10 (Non-vacuity)**.** A nonzero $P\in\mathcal{E}(f)\cap\mathcal{E}(\operatorname{rev} f)$ exists if and only if $D\neq 0$ (i.e., f is squarefree over $\mathbb{R}(\bar{x})$): "only if" is the gcd argument after Conjecture 2.2, and "if" is Remark 4.9. So the corrected hypothesis class is non-vacuous exactly under Brown's nondegeneracy condition. More is true: the intersection contains the product of any elements of the two factors, since each of $\mathcal{E}(f),\mathcal{E}(\operatorname{rev} f)$ is an ideal of R.

Remark 4.11 (The counterexample, revisited)**.** In Proposition 3.1, $\operatorname{rev} f=z^2+yz-xy^2$ is *monic*, so elimination is a closed projection and

$$V_{\mathbb{C}}(\mathcal{E}(\operatorname{rev} f))=\pi\big(V_{\mathbb{C}}(\operatorname{rev} f,(\operatorname{rev} f)')\big)=V\big(\operatorname{disc}(\operatorname{rev} f)\big)=V\big(y^2(1+4x)\big)\supseteq\{y=0\}.$$

Every element of $\mathcal{E}(\operatorname{rev} f)$ vanishes on the x-axis; $P=1+4x$ does not, so $P\notin\mathcal{E}(\operatorname{rev} f)$ and Theorem 4.8 correctly refuses the configuration.

5 Proof certificates for the pipeline

In the intended CAD proof-reconstruction pipeline, an external solver computes an element P of $\mathcal{E}(f)$ (typically the discriminant, via a subresultant computation) *together with* Bézout cofactors witnessing $P=Af+Bf_z$. Theorem 4.8 now demands a second certificate, for $\langle\operatorname{rev} f,(\operatorname{rev} f)'\rangle$. The next lemma shows the second certificate is a *mechanical transformation* of the first — no new solver call, no search — at the price of a factor $\operatorname{lc}(f)^M$.

Lemma 5.1 (Certificate transfer). *Let $f = \sum_{i=0}^{n} a_i z^i$ with $a_n \neq 0$, $n \geq 1$, and suppose*

$$P \;=\; Af + Bf_z, \qquad A, B \in R[z], \quad a := \deg A, \;\; b := \deg B, \quad P \in R.$$

Set $g := \operatorname{rev} f$, let $h := \operatorname{divX} g$ (so that $g = z\,h + a_n$, using that the constant coefficient of g is a_n), and let $M := \max(a+n,\; b+n-1)$. Then:

(i) (Reflection.) $z^M P \;=\; Ug + Vg'$, *where*

$$U \;=\; z^{\,M-n-a}\,\operatorname{rev}_a A \;+\; n\,z^{\,M-n+1-b}\,\operatorname{rev}_b B, \qquad V \;=\; -\,z^{\,M-n+2-b}\,\operatorname{rev}_b B.$$

This is obtained by applying rev_M to $P = Af + Bf_z$, using multiplicativity of rev *and Lemma 4.2 for the term $\operatorname{rev}_{n-1}(f_z) = ng - zg'$.*

(ii) (Clearing z^M.) *With $G := \sum_{j=0}^{M-1} \binom{M}{j} (-zh)^j\, g^{\,M-1-j}$, so that $a_n^M = (g - zh)^M = g\,G + (-zh)^M$,*

$$a_n^M\,P \;=\; \big(GP + (-h)^M U\big)\,g \;+\; \big((-h)^M V\big)\,g'.$$

Consequently $\operatorname{lc}(f)^M P \in \langle \operatorname{rev} f, (\operatorname{rev} f)' \rangle \cap R$, with explicit cofactors; and trivially $\operatorname{lc}(f)^M P = (a_n^M A) f + (a_n^M B) f_z \in \langle f, f_z \rangle$ as well. Both identities have been verified symbolically on the data of Proposition 3.1 ($n = 2$, $a = 1$, $b = 2$, $M = 3$).

Remark 5.2 (No transfer without the factor). Some correction factor is unavoidable: Proposition 3.1 exhibits $P \in \mathcal{E}(f) \setminus \mathcal{E}(\operatorname{rev} f)$, so no general implication $\mathcal{E}(f) \subseteq \mathcal{E}(\operatorname{rev} f)$ exists. The factor $\operatorname{lc}(f)^M$ is conceptually exactly right — it forces the witness to vanish on the degree-drop locus, which is the information $\mathcal{E}(f)$ lacks (Remark 3.2).

In the pipeline, the order-invariance of the corrected witness is free, because Brown's reduced projection already contains $\operatorname{lc} f$ as a projection factor, and projection factors are order-invariant on the cells of the induced CAD. This yields the form of the theorem most convenient for the algorithm:

Corollary 5.3 (Pipeline form). *Let f, $P \neq 0$, $P \in \mathcal{E}(f)$ be as above, and let S be a connected analytic submanifold on which* both P and $\operatorname{lc} f$ *are order-invariant, with $f(q, \cdot) \not\equiv 0$ for all $q \in S$. Then f is degree-invariant on S.*

Proof. On a connected S, order-invariance of $\operatorname{lc} f$ implies sign-invariance: if the constant order is 0 then $\operatorname{lc} f$ is nonvanishing on S, hence of constant sign by continuity and connectedness; if it is ≥ 1 then $\operatorname{lc} f \equiv 0$ on S. Let $\widetilde{P} := \operatorname{lc}(f)^M P$ with M as in Lemma 5.1; then $\widetilde{P} \neq 0$, $\widetilde{P} \in \mathcal{E}(f) \cap \mathcal{E}(\operatorname{rev} f)$, and $\widetilde{P}$ is order-invariant on S because orders add over products: $\operatorname{ord}_q \widetilde{P} = M \operatorname{ord}_q(\operatorname{lc} f) + \operatorname{ord}_q P$ is constant. Apply Theorem 4.8 with witness $\widetilde{P}$. □

Remark 5.4 (No new roots, no slowdown). The corrected witness $\widetilde{P} = \operatorname{lc}(f)^M P$ is never computed, expanded, factored, or evaluated by the CAD algorithm. The projection factor set remains

$$\{\operatorname{lc} F\} \;\cup\; \{\text{irreducible factors of } P(F)\} \;\cup\; \{\text{irreducible factors of } r(F, G)\},$$

identical to the naive generalized reduced projection, because $V(\widetilde{P}) = V(\operatorname{lc} f) \cup V(P)$ contributes no irreducible factor not already present. $\widetilde{P}$ exists only on the proof side, as the witness handed to Theorem 4.8; the exponent M is fixed once by the degrees of the cofactors A, B in the solver's certificate and appears only there. In particular the generalization causes no additional cells, no larger degrees, and no extra projection work. The pair elements $r(F, G) \in \langle F, G \rangle \cap R$ of the generalized projection are entirely unaffected: Brown's Theorem 3.1 concerns only the per-polynomial degree-invariance recovery.

6 Formalizing Theorem 3.1′ in Lean: an assessment

We recommend formalizing Corollary 5.3 *directly* ("Version A"), with witness construction via Lemma 5.1, rather than Theorem 4.8 ("Version B"). Version A avoids the bivariate homogenization layer entirely: in the pointwise argument one obtains the witness for f^*_γ from the chain

$$P \in \langle f, f_z \rangle \xrightarrow{\text{shift } z \mapsto z+\gamma} P \in \langle \bar{f}, \bar{f}_z \rangle \xrightarrow{\text{Lemma 5.1}} \mathrm{lc}(f)^M P \in \langle f^*_\gamma, (f^*_\gamma)' \rangle,$$

using $\mathrm{lc}\,\bar{f} = \mathrm{lc}\,f$, and order-invariance of the witness follows from the strengthened hypothesis exactly as in the proof of Corollary 5.3. Version A is also all the CAD pipeline needs, since cells of a McCallum-style CAD make every projection factor — including $\mathrm{lc}\,F$ — order-invariant.

Already available

- **The analytic core is done.** `lifting_theorem_generalized` (Theorem 1.1) is proved, axiom-free; it is invoked once, for f^* on $S \cap N$. No new analysis (Weierstrass preparation, monodromy, etc.) is required anywhere in the proof of Theorem 4.8.
- `AnalyticDelineable` already carries constant multiplicities (`rootMultiplicity (θ i a) = m i`), which is precisely what step (2) of the proof consumes.
- `IsAnalyticSubmanifold.straightening_chart` (Theorem 2.2.1) is proved, and the connectivity-refinement argument for $S \cap U$ already appears inline in `Generalized/Delineable.lean`; Lemma 4.7 is a repackaging.
- `order_invariant_mul_mv` / `order_invariant_pow_mv` give additivity of order over products, for the witness $\mathrm{lc}(f)^M P$.
- Mathlib supplies the entire reversal toolkit: `Polynomial.reflect`, `reflect_mul`, `coeff_zero_reverse`, `rootMultiplicity_eq_natTrailingDegree′` (mult_0 = trailing degree), `X_pow_dvd_iff`, `X_mul_divX_add` ($g = X \cdot \mathrm{divX}\, g + C(g(0))$), and `derivative_comp` (for the shift homomorphism $\mathrm{aeval}(X + C\,\gamma)$). Since `PolyR n` is literally `Polynomial (MvPolyR n)` and `specialize` is a `Polynomial.map`, specialization commutes with `reflect` definitionally (a one-line `ext`).

To build (Version A)

1. `SignInvariant` (new definition) and "order-invariant ⇒ sign-invariant on connected S" — small ($\approx$ 50–100 lines; the nonvanishing case uses connectedness and the intermediate value property, for which the repo has precedents).
2. Lemma 4.2 as a `reflect` identity — elementary coefficient computation ($\approx$ 50–100 lines).
3. Shift transport: `C P ∈ span {f, f′}` implies the same for $f(z+\gamma)$, via the algebra map $\mathrm{aeval}(X + C\gamma)$ and `derivative_comp` ($\approx$ 50 lines).
4. Lemma 5.1: `reflect_mul` + binomial expansion + `divX` bookkeeping. The statement to prove is

```
C P ∈ span {f, derivative f} →
C (lc f ^ M * P) ∈ span {reflect n f, derivative (reflect n f)}
```

with M existential ($\approx$ 200–350 lines; entirely computational, the proof in Section 5 is already phrased in Mathlib-shaped steps). Here `lc` abbreviates `leadingCoeff`, and the reversal is taken at the fixed formal degree n via `reflect`, which agrees with `reverse` since $a_n \neq 0$.

5. Lemma 4.7 as a standalone lemma from the straightening chart, plus the (easy) fact that `IsAnalyticSubmanifold` restricts to open subsets — the definition is pointwise-local with a fixed dimension, so the same local data restricted to $W \cap N$ works ($\approx$ 150–300 lines, mostly adapting the existing inline argument).

6. Trailing-degree bookkeeping ($\approx$ 100–150 lines): `rootMultiplicity_eq_natTrailingDegree'`, together with the commutation of `specialize` with `reflect` and with the shift, gives that the multiplicity of 0 as a root of `specialize (reflect n (shift f)) q` equals n minus `natDegree (specialize f q)`.

7. The main theorem `brown_3_1_generalized`: the pointwise argument (choose γ by finiteness of roots, choose N_0 by continuity of the polynomial ($q \mapsto f(q, \gamma)$), apply Theorem 1.1, identify and isolate the zero section, conclude), plus "locally constant $\Rightarrow$ constant" on a connected set ($\approx$ 300–500 lines; compare the similar clopen/branch-tracking arguments already in `Generalized/Delineable.lean`).

Total estimate: roughly 0.9–1.5k lines of new Lean, none of it deep — the difficulty profile is "elementary but fiddly polynomial algebra plus soft topology," comparable to `Projection.lean`-style glue, and far below the monodromy/Puiseux work already completed in this repository. The main uncertainty is item 5 (chart API friction) and the γ-selection topology in item 7.

Version B and integration

Version B (Theorem 4.8 verbatim, $\mathrm{lc}\, f$ only sign-invariant) additionally needs the homogenization layer of Lemma 4.4: a model of $R[Z, W]$ (e.g. `Polynomial (Polynomial (MvPolyR n))` or `MvPolynomial (Fin 2) (MvPolyR n)`), formal homogenization, the Euler relation, and the substitution automorphisms — all elementary, but an estimated additional 500–900 lines of API. It can be deferred: Version A is what the pipeline consumes.

Finally, integrating into a full "generalized reduced projection" theorem (the analogue of `mccallum_3_2_3_generalized` with the all-coefficients hypothesis `hcoeff` replaced by leading-coefficient order-invariance) is a moderate refactor: degree-invariance of each basis element now comes from `brown_3_1_generalized`, the hypothesis "$\forall q \in S$, `specialize f q` $\neq 0$" is assumed directly (as Brown does) instead of being derived from coefficient order-invariance, and the existing product/factor plumbing is reused unchanged (`degree_invariant_prod`, `delineable_factor_of_delineable_prod`, …; $\approx$ 100–200 lines).

References

[1] C. W. Brown, *Improved projection for cylindrical algebraic decomposition*, J. Symbolic Computation **32** (2001), 447–465.

[2] S. McCallum, *An improved projection operator for cylindrical algebraic decomposition*, in: Caviness, Johnson (eds.), *Quantifier Elimination and Cylindrical Algebraic Decomposition*, Springer, 1998.

[3] J.-P. Jouanolou, *Le formalisme du résultant*, Adv. Math. **90** (1991), 117–263.